HelsinkiTeam at HalluScoring-2026: Domain-Adaptive Arabic Hallucination Detection and Answer Selection with ALLaM

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Abstract

The rapid advancement of large language models (LLMs) has enabled powerful AI applications, yet hallucinations remain a major challenge to their reliability. As part of Track 2 of the HalluScoring 2026 shared task at ArabicNLP 2026, we present a system for detecting hallucinated answers and selecting the correct answer from six closely related candidates. We adapt the Arabic–English ALLaM-7B-Instruct model using Low-Rank Adaptation (LoRA) with domain-specific supervision from Qur'anic text, Hadith, Islamic law and inheritance, Islamic history, and general knowledge. The system jointly performs hallucination detection and answer selection, achieving 88.9% accuracy and 88.6% macro-F1 for hallucination detection, and 91.1% answer selection accuracy on the 800-instance development set. Error analysis identifies geography and structured-output parsing as the main remaining weaknesses.

1 Introduction

Over the last decade, natural language processing (NLP) has advanced rapidly with the emergence of Transformer-based large language models (LLMs) (Vaswani et al., 2017; Bounab et al., 2025). Arabic has received increasing attention due to the growing demand for robust language technologies and its unique linguistic characteristics (Bekhouche et al., 2025). Despite their ability to generate fluent Arabic text, multilingual and Arabic-specific LLMs remain susceptible to hallucinations, including unsupported names, dates, quotations, and factual claims. Such errors are particularly consequential in religious and legal contexts, where minor lexical changes can alter the meaning of a Qur'anic verse, Hadith, or inheritance ruling. Detection is further complicated by Arabic's rich morphology, spelling variations, and culturally specific knowledge.

HalluScoring 2026 addresses these challenges through a benchmark for hallucination detection, factuality verification, and trustworthy Arabic question answering (Alansari et al., 2026). Track 2 requires systems to determine whether a generated answer is hallucinated and select the correct answer from six closely related candidates. It comprises Task 2.1 on Islamic knowledge and Task 2.2 covering geography, science, history, and Islamic knowledge.

These tasks highlight the broader need for trustworthy LLMs, particularly in high-stakes applications where plausible but incorrect information can have serious consequences. Motivated by this need, this work makes the following contributions:

- A domain-adaptive training strategy that combines canonical Islamic material, controlled corruptions, inheritance reasoning, and broad synthetic factuality examples.
- A unified framework for jointly performing hallucination detection, candidate answer selection, and explanation generation.
- A reproducible evaluation on the HalluScoring 2026 development set, including domain-level and structured error analyses.

2 Related Work

Hallucinations remain a major challenge to the reliability of large language models (LLMs), motivating research on their characterization, detection, and mitigation. They are commonly categorized by factuality and faithfulness, or as intrinsic and extrinsic errors (Kumar, 2025; Cleti and Jano, 2024; Mohammadabadi et al., 2025; Pesaranghader and Li, 2026). Detection approaches include uncertainty and internal-state analysis (Hou et al., 2025; Aisin and Shamardina, 2025; Wang et al., 2025a), self-consistency and multi-agent reasoning (Friel and Sanyal, 2023; Sun et al., 2025), and external fact verification (Wang et al., 2025b; Su et al., 2025). Multilingual and domain-specific studies further

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show that hallucination behavior varies across languages and applications (Islam et al., 2025; Joshi, 2025; Spracklen et al., 2025).

Arabic hallucination resources remain relatively limited. HalluScore (Alansari and Luqman, 2026) provides 827 curated Arabic questions with evidence, explanations, and human annotations of responses from 17 LLMs. HalluScoring extends this setting by jointly evaluating hallucination detection and selection among closely related candidate answers. MAWARITH (Bouchekif et al., 2026b) provides structured Arabic inheritance cases and intermediate legal decisions, which we use for domain-adaptive supervision while targeting the distinct task of factual hallucination detection and answer selection.

For efficient adaptation, LoRA (Hu et al., 2022) introduces trainable low-rank updates while keeping pretrained model weights frozen, reducing the cost of full fine-tuning. We apply LoRA to ALLaM-7B with a domain-adaptive Arabic corpus containing authentic and synthetic hallucination examples, formulating hallucination detection and answer selection as a unified instruction-following generation task for HalluScoring 2026.

3 Task Definition

Participants must jointly determine whether a generated answer is hallucinated and select the correct answer from six candidates. Each instance consists of a question q, generated answer g, optional reference r, and candidates $O = \{o_A, \ldots, o_F\}$. The system predicts a hallucination label $y \in \{\text{HALLUCINATION}, \text{NO-HALLUCINATION}\}$ and the correct candidate $z \in \{A, \ldots, F\}$. It additionally generates a justification and identifies erroneous spans for hallucinated answers, although only the label and candidate selection are used for quantitative evaluation.

Since these tasks are related but distinct, a correct candidate may be selected even when the hallucination label is incorrect, and vice versa. We therefore report label accuracy, macro-F1, candidate selection accuracy, and strict joint accuracy, which requires both predictions to be correct.

4 Benchmark & Domain-Adaptive Data

We evaluate our approach on **HalluTruthQA-4K**, the official Subtask 2.2 dataset of HalluScoring 2026, comprising 4,000 expert-annotated Arabic instances equally distributed across Islamic knowledge, history, science, and geography. Each instance contains a binary hallucination label and six candidate answers with one verified correct option, while hallucinated instances additionally include erroneous spans and human-written explanations. The official split contains 1,600 training, 800 development, and 1,600 held-out test instances (Bouchekif et al., 2026c; Bekhouche et al., 2026).

We further enrich the training data with approximately 30,000 domain-specific examples after filtering and deduplication, extending coverage of Islamic knowledge and fine-grained factual corruptions. As some source groups overlap after balancing, the component counts reported below do not represent unique examples.

- **Controlled hallucinations:** We generated 16,000 negative examples using localized corruptions, including character or word replacement, deletion, sentence modification, and changes to Surah names or verse numbers, exposing the model to subtle factual inconsistencies.
- **Canonical religious texts:** We include 8,000 Qur'anic and 4,000 Hadith examples, pairing authentic content with carefully constructed corrupted alternatives.
- **Inheritance and Islamic law:** We use 4,000 examples derived from MAWARITH (Bouchekif et al., 2026b), introduced for the QIAS 2026 shared task (Bouchekif et al., 2026a). ALLaM-generated answers were compared with annotated references, and discrepancies were converted into supervised hallucination examples.
- **Broader synthetic coverage:** We generated 17,000 candidate examples covering Fiqh, Islamic history, the biography of the Prophet Muhammad, and related topics.

For each question in the adaptation corpus, we generated both a factually supported answer and an intentionally hallucinated alternative. After format validation and quality filtering, the retained examples were incorporated into training. This diverse mixture exposes the model to both fine-grained textual corruptions and higher-level knowledge contradictions, while the inclusion of both positive and negative examples helps reduce bias toward classifying domain-specific statements as hallucinated.

5 System Description

5.1 Base model & Parameter-Efficient Fine-Tuning

The proposed system uses ALLaM-7B-Instruct-preview[1], an Arabic–English autoregressive Transformer (Bari et al., 2025) with 7B parameters, 32 decoder layers, a 4,096-dimensional hidden size, 32 attention heads, a 64K-token vocabulary, and a 4,096-token context length. Its bilingual pretraining and Arabic instruction tuning make it well suited for Arabic factuality verification.

We freeze the base model and apply LoRA to the attention and feed-forward projections using $r = 32$, $\alpha = 64$, dropout of 0.05, and no bias. This adds approximately 80M trainable parameters (1.1% of the model). Table 1 summarizes the fine-tuning configuration.

Hyperparameter	Value
Epochs	3
Per-device batch size	1
Gradient accumulation	16
Effective batch size	16
Learning rate	2×10^{-5}
Weight decay	0.01
Warm-up ratio	0.03
Scheduler	Cosine
Maximum sequence length	4,096
Precision	BF16
Gradient checkpointing	Enabled
Distributed optimization	DeepSpeed
Training objective	Completion-only LM loss
LoRA rank / alpha	32 / 64

Table 1: LoRA Fine-Tuning Configuration.

Completion-only loss masks the system and user tokens in each training sequence, so the loss is computed only over the target assistant response. This encourages schema compliance while preventing the model from learning to reproduce the input prompt.

5.2 Inference Pipeline

After fine-tuning, we load the frozen ALLaM checkpoint in BF16 on a single GPU and attach the PEFT adapter. The model is prompted through its chat template with an Arabic system message defining three tasks: hallucination detection, correct-option selection, and erroneous-span identification. The user message contains the question, generated answer, available gold reference, and six options (A–F). The model returns a JSON object with `predicted_label`, `predicted_answer`, `selected_option`, `justification`, and `hallucinations`.

Inference uses greedy decoding (`do_sample=False`) with a maximum of 384 generated tokens and processes one instance at a time. During post-processing, we extract the first JSON object, normalize the predicted label, and retrieve the selected option from `selected_option`, `predicted_answer`, or `pred_option`, in that order. Empty `span_start` or `span_end` values are replaced with `null` before JSON decoding, as these offsets are not required for evaluation.

6 Development Results & Error Analysis

6.1 Results

We evaluate on the 800-instance Task 2.2 development set, generated by `QCRI/Fanar-1-9B-Instruct`, with 200 instances each from Islamic knowledge, history, science, and geography. The set contains 359 hallucinated and 441 non-hallucinated answers.

All metrics include format failures; parsing exceptions are assigned HALLUCINATION with an empty answer option. Macro-F1 is the unweighted mean of both class F1 scores. Table 2 reports these local development results, which are not official test-set scores.

Metric	Score (%)
Hallucination-label accuracy	88.9
Label macro-F1	88.6
Correct-option accuracy	91.1
Strict joint accuracy	84.0
Valid structured outputs	96.5

Table 2: Results on the 800-instance Task 2.2 development set.

Table 2 shows that candidate selection is more accurate than binary hallucination detection. The label confusion matrix comprises 291 true positives, 420 true negatives, 21 false positives, and 68 false negatives. For the hallucination class, the model achieves a precision of 93.3%, a recall of 81.1%, and an F1 score of 86.7%. These results indicate that the model is conservative: hallucination predictions are usually correct, but some subtle or fine-grained hallucinations remain undetected.

[1] `humain-ai/ALLaM-7B-Instruct-preview`

To better understand the model's behavior across knowledge areas, we further analyze the development results by domain. Table 3 reports the label and answer-option accuracies for each of the four Task 2.2 domains.

Domain	Label acc.	Option acc.
Islamic knowledge	93.0	95.5
History	90.5	88.5
Science	93.0	99.0
Geography	79.0	81.5

Table 3: Development accuracy (%) by domain; each row contains 200 instances.

The results show clear domain differences. Geography is the most challenging, with the lowest label and option accuracies, while science achieves the highest option accuracy (99.0%). History is the only domain where label accuracy exceeds option accuracy, suggesting greater difficulty distinguishing among similar candidate answers.

6.2 Error Analysis

We qualitatively examine the main failure modes, focusing on subtle lexical differences, historical ambiguity, structured generation failures, and reference-based evaluation.

Small lexical substitutions. Difficult false negatives often involve a single incorrect name or character within an otherwise correct answer. For example, the model may produce the Tunisian *al-Qayrawan* instead of the Moroccan university *al-Qarawiyyin*, where high semantic similarity can obscure the localized factual error.

Ambiguous historical attribution. Some questions allow multiple interpretations. Muhammad ibn Tumart may be regarded as the spiritual founder of the Almohad movement, whereas Abd al-Mu'min established and expanded the ruling state. A single gold answer may therefore penalize otherwise defensible responses.

Structured generation failures. Twenty-eight of 800 outputs (3.5%) caused parsing exceptions, commonly due to missing character-offset values. Replacing these with `null` resolves some cases, while constrained decoding could further reduce such failures.

Reference leakage and realism. The development prompt includes reference answers when available, making the setting closer to reference-based verification than closed-book detection. The results therefore should not be interpreted as reference-free factual recall; an ablation without references is needed to assess this capability.

7 Conclusion

Large language models have advanced Arabic question answering, yet hallucinations remain a critical challenge. We presented a LoRA-adapted ALLaM-7B system for HalluScoring 2026, jointly addressing hallucination detection and answer selection through domain-adaptive supervision from authentic and synthetic Arabic data. The system achieved strong local development performance, particularly in Islamic knowledge and science, while geography, historical ambiguity, lexical substitutions, and structured output failures remain key challenges.

Future work will explore reference-free evaluation, constrained decoding, authoritative knowledge retrieval, cross-model validation, and systematic ablations of the training data to improve the robustness and reliability of Arabic hallucination detection.

Ethical Considerations

Religious texts and Islamic inheritance law are sensitive domains. This system is intended for benchmark research and error detection, not for issuing fatwas or determining legal shares. Synthetic corruptions should remain clearly separated from canonical sources to prevent their misuse as authentic material. Deployment should involve qualified Arabic-speaking domain experts and clearly communicate model uncertainty and supporting evidence to users.

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[2] https://sites.utu.fi/lifefactfuture/

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